

Expanded polyglutamine stretches interact with TAF_{II}130, interfering with CREB-dependent transcription

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At least eight inherited neurodegenerative diseases are caused by expanded CAG repeats encoding polyglutamine (polyQ) stretches. Although cytotoxicities of expanded polyQ stretches are implicated, the molecular mechanisms of neurodegeneration remain unclear. We found that expanded polyQ stretches preferentially bind to TAF_{II}130, a coactivator involved in cAMP-responsive element binding protein (CREB)-dependent transcriptional activation, and strongly suppress CREB-dependent transcriptional activation. The suppression of CREB-dependent transcription and the cell death induced by polyQ stretches were restored by the co-expression of TAF_{II}130. Our results indicate that interference of transcription by the binding of TAF_{II}130 with expanded polyQ stretches is involved in the pathogenetic mechanisms underlying neurodegeneration.

Expansions of CAG repeats coding for polyQ stretches have been found to be the causative mutation in at least eight neurodegenerative disorders, including spinal and bulbar muscular atrophy (SBMA), Huntington disease (HD), dentatorubral-pallidoluysian atrophy (DRPLA), spinocerebellar ataxia type 1 (SCA1), SCA2, Machado-Joseph disease (MJD, also called SCA3), SCA6 and SCA7 (ref. 1). For each of these diseases there is a threshold repeat length beyond which individuals become affected. Generally, greater than 35–40 CAG repeats are associated with diseases, except SCA6, for which the threshold length is 20 repeats². There is an inverse correlation between the size of the expanded CAG repeats and the age at onset, suggesting that the length of the polyQ stretches is related to the pathogenetic mechanisms underlying these diseases.

Recent studies have demonstrated that the expression of truncated proteins containing the expanded polyQ stretches results in the formation of aggregate bodies and apoptosis in transient expression systems. Transgenic mice expressing proteins containing these expanded polyQ stretches exhibit neurological phenotypes^{3,4}. Neuronal intranuclear inclusions (NIIs) containing mutant proteins have been demonstrated in HD- and SCA1-transgenic mice as well as in autopsied brains of these CAG-repeat diseases^{1,5–12}.

Although the physiological functions of the mutant proteins in these diseases remain unclear (except for the androgen receptor, whose gene is mutated in SBMA), wild-type ataxin-1, ataxin-7, MJD1a and DRPLA proteins as well as the androgen receptor have been localized predominantly in the nuclei^{5,13–17}. Such nuclear localization is presumably mediated by putative nuclear localiza-

tion signals (NLS) in these proteins. In the case of huntingtin protein, an amino-terminal fragment containing an expanded polyQ stretch has been demonstrated to be translocated to the nucleus^{7,18}. The importance of nuclear translocation of mutant proteins has been further demonstrated by the observation that transgenic mice expressing mutant ataxin-1 with a mutated NLS did not develop ataxia¹⁹, and that the addition of a nuclear export signal (NES) to mutant huntingtin suppressed the formation of NIIs and apoptosis²⁰.

Given that the nuclear translocation of mutant proteins is a prerequisite for their cytotoxicity, we hypothesized that nuclear proteins that interact with the mutant proteins, in particular those that bind to the expanded polyQ stretches, may be involved in the pathogenetic mechanisms underlying neurodegeneration. We found that expanded polyQ stretches bind to a human TATA-binding protein (TBP)-associated factor, human TAF_{II}130, a coactivator involved in cAMP-responsive element-binding protein (CREB)-dependent transcriptional activation^{21,22}, and inhibit CREB-dependent transcriptional activation. We propose that interference with transcriptional activation due to the binding of the expanded polyQ stretches to TAF_{II}130 is involved in the pathogenetic mechanisms underlying CAG-repeat diseases.

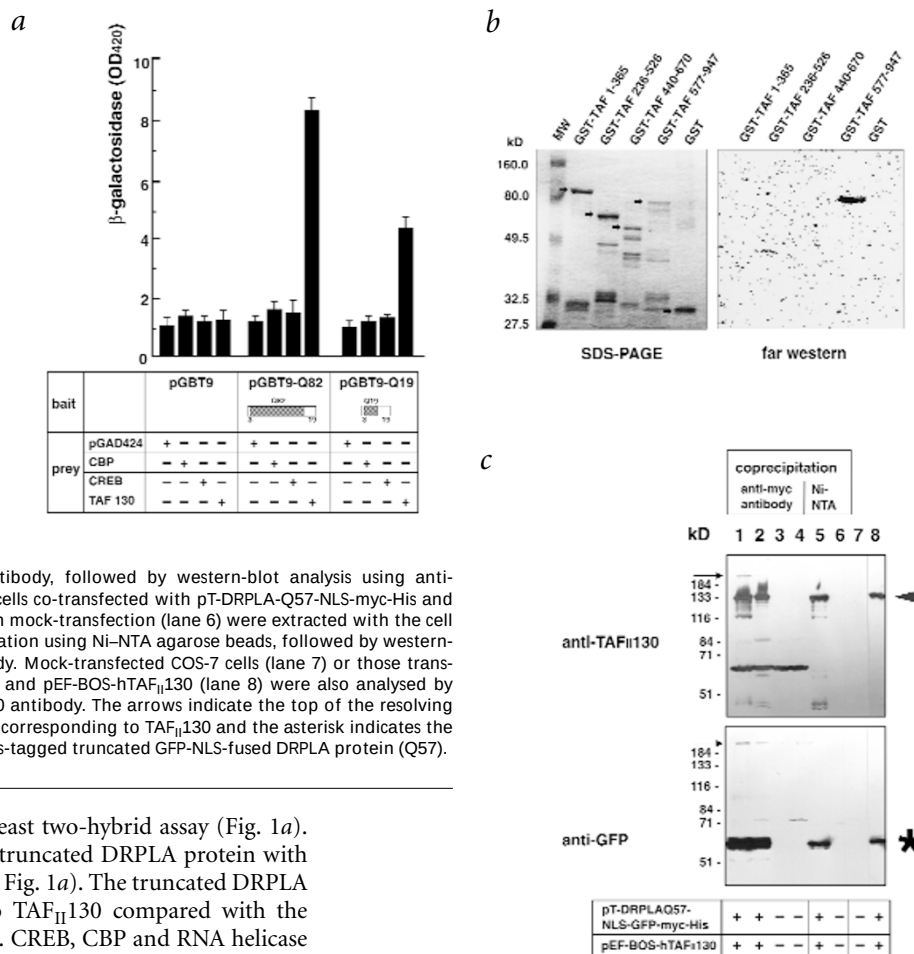
Results

TAF_{II}130 interacts with polyQ stretches

We examined nuclear proteins, including CREB, CREB-binding protein (CBP), RNA helicase A and TAF_{II}130, for their ability to

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Fig. 1 Interaction between TAF_{II}130 and polyQ stretches was demonstrated by the yeast two-hybrid assay (a), the far western assay (b) and the co-precipitation assay (c). **a**, DNA segments encoding truncated DRPLA protein with polyQ stretches (Q19 or Q82) were subcloned into the pGBT9 bait vector (pGBT9-Q19 or pGBT9-Q82). DNA segments coding for full-length CREB, CBP, RNA helicase A or TAF_{II}130 were subcloned into pGAD424 prey vectors. Results are mean $\pm$ s.d. (n=3). **b**, Far-western blot of GST-TAF_{II}130 fusion proteins containing various regions of TAF_{II}130 (aa 1–365, 236–526, 440–670 and 577–947; 0.5 μ g each). **c**, COS-7 cells were transfected with pT-DRPLA-Q57-NLS-myc-His encoding myc- and His-tagged GFP-NLS-fused truncated DRPLA protein (Q57) and pEF-BOS-hTAF_{II}130 (lanes 1, 2), or mock-transfected (lanes 3, 4). The nuclear extracts obtained using the lysis buffer containing 1% NP40 (lanes 1, 3) or buffer C (lanes 2, 4) were immunoprecipitated with anti-myc antibody, followed by western-blot analysis using anti-TAF_{II}130 antibody. The nuclei of COS-7 cells co-transfected with pT-DRPLA-Q57-NLS-myc-His and pEF-BOS-hTAF_{II}130 (lane 5) or those with mock-transfection (lane 6) were extracted with the cell lysis buffer, and subjected to co-precipitation using Ni-NTA agarose beads, followed by western-blot analysis using anti-TAF_{II}130 antibody. Mock-transfected COS-7 cells (lane 7) or those transfected with pT-DRPLA-Q57-NLS-myc-His and pEF-BOS-hTAF_{II}130 (lane 8) were also analysed by western-blot analysis using anti-TAF_{II}130 antibody. The arrows indicate the top of the resolving gel. The arrowhead indicates the bands corresponding to TAF_{II}130 and the asterisk indicates the bands corresponding to the myc- and His-tagged truncated GFP-NLS-fused DRPLA protein (Q57).



bind to polyQ stretches by the yeast two-hybrid assay (Fig. 1a). TAF_{II}130 strongly bound to the truncated DRPLA protein with an expanded polyQ stretch (Q82; Fig. 1a). The truncated DRPLA protein (Q19) bound weakly to TAF_{II}130 compared with the truncated DRPLA protein (Q82). CREB, CBP and RNA helicase A did not bind to either of the truncated DRPLA proteins.

To define the polyQ-binding domain in TAF_{II}130, we evaluated the binding between GST-TAF_{II}130 fusion proteins carrying a subset of truncated TAF_{II}130 (aa 1–365, 236–526, 440–670 and 577–947) and expanded polyQ stretches (Q82) using the far-western procedure. The carboxy-terminal one-fourth of TAF_{II}130 (residues 577–947) was the only protein that bound to the ³⁵S-labelled polyQ probe (Q82; Fig. 1b), indicating that the C-terminal amino acids 670–947 were involved in the interaction with expanded polyQ stretches. The interaction of expanded polyQ stretches with amino acids 670–947 of TAF_{II}130 was also confirmed by the yeast two-hybrid assay (data not shown).

Co-precipitation of TAF_{II}130 and expanded polyQ stretches

To investigate whether expanded polyQ stretches interact with TAF_{II}130 in cultured cells, we expressed the myc- and His-tagged truncated GFP-NLS-fused DRPLA protein (Q57) and TAF_{II}130 in COS-7 cells, and performed co-precipitation studies. The myc- and His-tagged truncated DRPLA protein (Q57) in the nuclear extract was immunoprecipitated with the anti-myc antibody, and then subjected to western-blot analysis using the anti-TAF_{II}130 monoclonal antibody. The interaction between TAF_{II}130 and expanded polyQ stretches was clearly demonstrated (Fig. 1c, lanes 1–4). A faint band detected by the anti-TAF_{II}130 monoclonal antibody was observed at the gel top (Fig. 1c, lane 1), suggesting that TAF_{II}130 is in part incorporated into the aggregates of expanded polyQ stretches. The interaction between expanded polyQ stretches and TAF_{II}130 was further confirmed by the co-precipitation assay using nickel-nitriloacetic acid (Ni-NTA) agarose beads (Fig. 1c, lanes 5, 6).

Co-localization of TAF_{II}130 with aggregate bodies

We then investigated whether TAF_{II}130 is co-localized with aggregate bodies in transient expression studies. In COS-7 cells expressing truncated DRPLA protein (Q57), distinct aggregate bodies were formed mainly in the cytoplasm (Fig. 2s). Co-expression of pT-DRPLAQ56-GFP and pEF-BOS-hTAF_{II}130 demonstrated that TAF_{II}130 was co-localized with the aggregate bodies (Fig. 2a–c), whereas expression of TAF_{II}130 alone (pEF-BOS-hTAF_{II}130) showed diffuse staining in the nuclei (Fig. 2u).

To confirm that the expanded polyQ stretches, and not the flanking sequences, are involved in the co-localization of TAF_{II}130 with the aggregate bodies, we carried out similar experiments for truncated mutant huntingtin (Q81), ataxin-2 (Q64) or MJD1a protein (Q72). Co-expression of these proteins with TAF_{II}130 similarly resulted in the formation of cytoplasmic aggregate bodies, and co-localization of TAF_{II}130 with these aggregate bodies was shown (Fig. 2d–l). Only the rims of the aggregate bodies were stained with anti-TAF_{II}130 antibody (Fig. 2b,e,h). Despite the intense and homogeneous fluorescence of green fluorescence protein (GFP) in the aggregate bodies (Fig. 2a,d,g), only the rim of the aggregate bodies was stained with anti-GFP antibody (data not shown), suggesting limited accessibility of the antibodies to the core of the aggregate bodies.

In the above experiments, aggregate bodies were formed mainly in the cytoplasm, because the putative NLS sequences had been removed in the truncated mutant proteins for MJD/SCA3 and DRPLA. Because huntingtin, ataxin-1, MJD1a and DRPLA proteins are translocated into the nucleus^{5,7,14–16,19,23}, we considered that an experimental design that allows the formation of intranu-

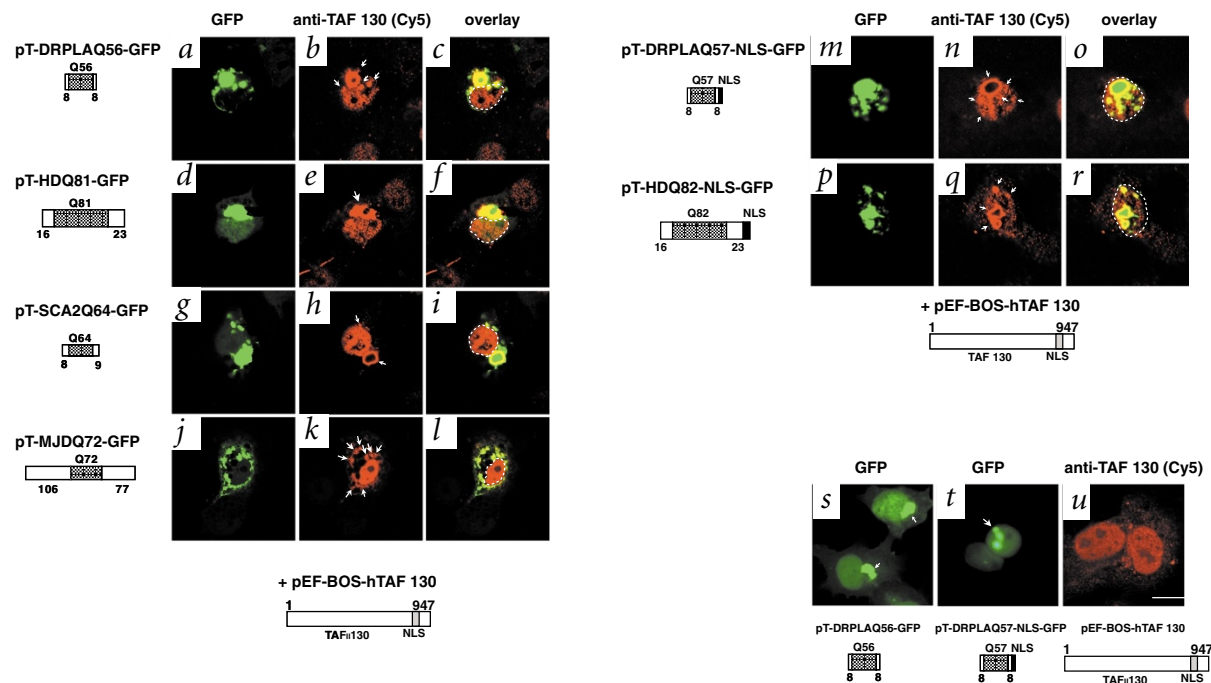


Fig. 2 TAF_{II}130 co-localizes with aggregate bodies in transient expression systems. **a–l**, Co-localization of TAF_{II}130 with cytoplasmic aggregate bodies. COS-7 cells were co-transfected with pEF-BOS-hTAF_{II}130 and one of the following plasmids; pT-DRPLAQ56-GFP (**a–c**), pT-HDQ81-GFP encoding truncated huntingtin (Q81) (**d–f**), pT-SCA2Q64-GFP encoding truncated ataxin-2 (Q64; **g–i**), or pT-MJDQ72-GFP encoding truncated MJD1a protein (Q72; **j–l**). The cells were fixed 48 h after transfection and observed by GFP fluorescence (**a,d,g,j**) or by immunostaining using an anti-TAF_{II}130 antibody (**b,e,h,k**). The overlaid images are shown in (**c,f,i,l**). The frequencies of the cells exhibiting immunoreactivity for TAF_{II}130, which is co-localized with cytoplasmic aggregate bodies (indicated by arrowheads) containing truncated mutant DRPLA protein (Q56), huntingtin (Q81), ataxin-2 (Q64) or MJD1a protein (Q72), are 20% (10/50), 26% (13/50), 32% (16/50) and 22% (11/50), respectively. The nuclei as detected by Hoechst 33342 staining are shown by dotted lines in the overlaid images. The frequencies of the cells exhibiting immunoreactivity for TAF_{II}130 with intranuclear aggregate bodies (shown by arrowheads) containing truncated mutant DRPLA protein (Q57) or huntingtin (Q82), are 64% (32/50) and 58% (29/50), respectively. The nuclei as detected by Hoechst 33342 staining are shown by dotted lines in the overlaid images. **m–r**, Co-localization of TAF_{II}130 with intranuclear aggregate bodies (shown by arrowheads) containing truncated mutant DRPLA protein (Q57) or huntingtin (Q82), are 64% (32/50) and 58% (29/50), respectively. The nuclei as detected by Hoechst 33342 staining are shown by dotted lines in the overlaid images. **s–u**, Expression patterns of pT-DRPLAQ56-GFP encoding truncated DRPLA protein (Q56) (**s**), pT-DRPLAQ57-NLS-GFP encoding truncated DRPLA protein (Q57) with an NLS (**t**) or pEF-BOS-hTAF_{II}130 (**u**) in COS-7 cells. Cytoplasmic (**s**) and intranuclear aggregate bodies (**t**) are indicated by arrows. The cells were fixed 48 h after transfection and observed by GFP fluorescence (**s,t**) or by immunostaining using an anti-TAF_{II}130 antibody (**u**). Scale bar, 10 μ m.

clear aggregates of expanded polyQ stretches would be more appropriate for investigation of the co-localization of TAF_{II}130 with these aggregate bodies. We tested this possibility using pT-DRPLAQ57-NLS-GFP, encoding the truncated mutant DRPLA protein (Q57) along with an NLS (ref. 24). When COS-7 cells were transfected with pT-DRPLAQ57-NLS-GFP, a strong GFP fluorescence was observed only in the nuclei, and several large intranuclear aggregates were formed (Fig. 2*t*). In COS-7 cells co-expressing truncated DRPLA protein (p-DRPLAQ57-NLS-GFP) and TAF_{II}130, most of the intranuclear aggregate bodies were strongly stained with anti-TAF_{II}130 antibody (Fig. 2*m–o*). Such co-localization was also observed in COS-7 cells co-expressing truncated mutant huntingtin (pT-HDQ82-NLS-GFP; Fig. 2*p–r*).

In contrast to the diffuse staining of the nuclei with anti-TAF_{II}130 antibody in cells expressing TAF_{II}130 alone (Fig. 2*u*), TAF_{II}130 became sequestered into intranuclear aggregate bodies in cells co-expressing expanded polyQ stretches and TAF_{II}130 (Fig. 2*n,q*).

Co-localization of TAF_{II}130 and NIIs in autopsied brains of DRPLA and MJD/SCA3 cases

We then carried out immunohistochemical examination of autopsied brains of DRPLA and MJD/SCA3 cases to investigate whether TAF_{II}130 is co-localized with NIIs. Strong immunoreactivity for TAF_{II}130 was found in the NIIs in all the brains examined (Fig. 3). The NIIs in all the cases examined also showed weak to moderate immunoreactivity for CREB (Fig. 4*a,b*). Immunoreactivity for Sp1 was also found in the NIIs in both DRPLA (Fig. 4*c*) and MJD/SCA3 (Fig. 4*d*) cases. Weak immunoreactivity for TBP was also found in the NIIs in cases of DRPLA (Fig. 4*e*) and MJD/SCA3 (Fig. 4*f*). Other transcription factors, such as C/EBP- α and β , were not co-localized with NIIs (data not shown).

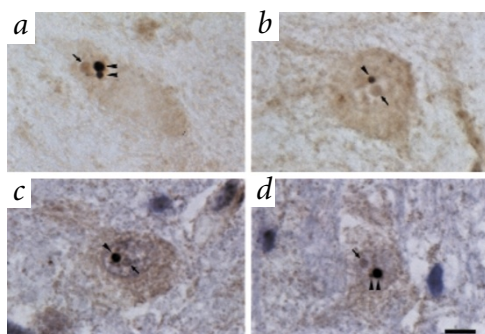


Fig. 3 Co-localization of TAF_{II}130 with NIIs in the autopsied brains of DRPLA and MJD/SCA3 cases. NIIs in the cerebellar dentate nuclei of DRPLA cases (**a,b**) and those in the pontine nuclei of a MJD/SCA3 case (**c,d**) show intense immunoreactivity for TAF_{II}130 (arrowheads). The frequencies of the neurons exhibiting NIIs co-localized with TAF_{II}130-immunoreactivity in autopsied brains of DRPLA and MJD/SCA3 cases are 83% and 60%, respectively. Nucleoli (arrows) were negative for TAF_{II}130. Scale bar, 10 μ m.

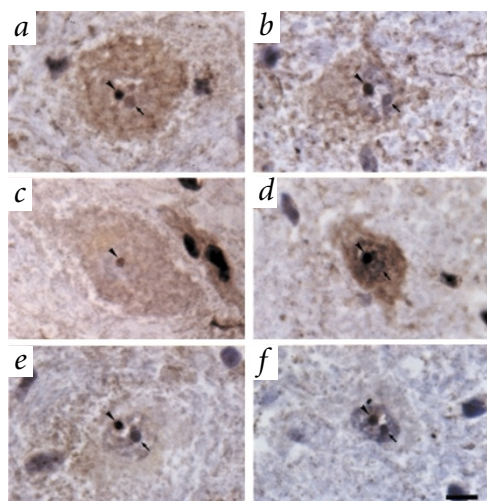


Fig. 4 Co-localization of Nils with the transcription factors CREB, Sp1 and TBP in the autopsied brains of DRPLA and MJD/SCA3 cases. Immunoreactivities for CREB (**a,b**), Sp1 (**c,d**) and TBP (**e,f**) are noted in the Nils in the cerebellar dentate nuclei of DRPLA cases (**a,c,e**) and in the pontine nuclei of a MJD/SCA3 case (**b,d,f**) (arrowheads). The frequencies of the neurons exhibiting immunoreactivity for CREB, Sp1 or TBP in DRPLA brains are 65%, 53% and 43%, respectively. The frequencies of the neurons exhibiting Nils with immunoreactivity for CREB, Sp1 or TBP in MJD/SCA3 are 46%, 52% and 30%, respectively. Nucleoli (arrows) were negative for these antibodies. Scale bar, 10 μ m.

Suppression of CREB-dependent transcriptional activation by expanded polyQ stretches

Because TAF_{II}130 is a transcriptional coactivator for CREB (refs 21,25), the binding of expanded polyQ stretches to TAF_{II}130 may interfere with the CREB-dependent transcriptional activation. To investigate this possibility, 293 cells were co-transfected with a GAL4-CREB expression vector and a G5B-luciferase reporter vector (G5B-Luc) containing five GAL4 recognition sites. Luciferase activity increased 7.3-fold in response to forskolin, an activator of protein kinase A (PKA) pathway, presumably through phosphorylation of CREB (Fig. 5). When the 293 cells were co-transfected with pEF-BOS-FQ82 coding for polyQ stretches (Q82), the forskolin-induced increase in luciferase activity was substantially suppressed (Fig. 5), whereas only slight suppression was observed for those co-transfected with pEF-BOS-FQ19.

We then investigated whether co-expression of TAF_{II}130 in addition to polyQ stretches restored this suppression of CREB-dependent transcriptional activation. When 293 cells were co-transfected with GAL4-CREB and wild-type PKA vectors, the luciferase activity increased 4.8-fold compared with that in cells co-transfected with GAL4-CREB and catalytically inactive (mutant) PKA vectors (Fig. 6a, lanes 1, 2). Co-expression with polyQ stretches (Q82) suppressed this increase in the luciferase activity in a manner dependent on the dose of pEF-BOS-FQ82 (Fig. 6a, lanes 3–5). Co-expression with TAF_{II}130 (pBK-CMV-hTAF_{II}130) restored the luciferase activity suppressed by polyQ stretches (Q82) to levels comparable to those observed in the absence of expanded polyQ stretches (Q82) (Fig. 6a, lanes 6–8).

To further confirm the involvement of TAF_{II}130 in the suppression of transcriptional activation by expanded polyQ stretches, we used the Q2 domain of CREB fused to GAL4. The GAL-Q2 protein constitutively activates the transcription of the GAL4-reporter gene without its prior phosphorylation²⁶. In this simplified system, expanded polyQ stretches (Q82) strongly sup-

pressed the transcriptional activation of the GAL4-luciferase reporter gene in a manner dependent on the dose of pEF-BOS-FQ82 (Fig. 6b, lanes 5–7). Co-expression of TAF_{II}130 (pBK-CMV-hTAF_{II}130) restored the luciferase activity suppressed by polyQ stretches (Q82) to levels comparable to those in the absence of expanded polyQ stretches (Fig. 6b, lanes 8–10).

TAF_{II}130 suppresses cell death caused by expanded polyQ stretches

To determine whether the binding of expanded polyQ stretches to TAF_{II}130 is involved in the cytotoxicity, the effect of co-expression of TAF_{II}130 and expanded polyQ stretches on COS-7 cells with respect to suppression of cell death was investigated. The cells expressing the truncated DRPLA protein (Q57) exhibited a high frequency of nuclear fragmentation compared with those expressing the truncated wild-type DRPLA protein (Q19). When the cells were co-transfected with pEF-BOS-hTAF_{II}130 and pT-DRPLAQ57-NLS-GFP, however, cell death was significantly reduced from 39.3 $\pm$ 5.5% to 26.7 $\pm$ 3.2% (P <0.05; Fig. 7a). In contrast, in cells co-transfected with pT-DRPLAQ57-NLS-GFP and pEF-BOS-hTAF_{II}130-N coding for the N-terminal three-fourths of TAF_{II}130 lacking the polyQ-binding domain, there was no reduction in the percentage of cell death. Co-transfection with pEF-BOS-hTAF_{II}130 and pT-DRPLAQ19-NLS-GFP did not show such an effect, indicating that the suppressive effect of pEF-BOS-hTAF_{II}130 is specific for expanded polyQ stretches.

A high frequency of aggregate body formation was observed in the cells co-transfected with pT-DRPLAQ57-NLS-GFP, whereas no such aggregate body formation was observed in the cells transfected with pT-DRPLAQ19-NLS-GFP (Fig. 7b). Co-expression with TAF_{II}130 only slightly reduced the rate of aggregate body formation (Fig. 7b). Co-expression with TAF_{II}130-N did not reduce the rate of aggregate body formation (Fig. 7b).

To investigate whether the binding of expanded polyQ stretches to TAF_{II}130 is involved in the neuronal cell death induced by expanded polyQ stretches, we conducted similar experiments using neuro2a cells, a mouse neuroblastoma cell line. Neuro2a cells exhibited similarly high vulnerability (41.0 $\pm$ 3.4%) to

Fig. 5 Expanded polyQ stretches inhibit CREB-dependent transcriptional activation. A G5B-luciferase reporter plasmid containing five GAL4 recognition sites was co-transfected into 293 cells along with a GAL4-CREB expression vector, RSV- β gal control plasmid, and/or polyQ expression plasmid (pEF-BOS-FQ19 or pEF-BOS-FQ82). Six hours after addition of the DNA/liposome mixture, forskolin dissolved in DMSO or DMSO alone was added into the culture medium and the culture was continued for another 24 h. Luciferase activity was normalized to the β -galactosidase activity, and the relative luciferase activity of the cells co-transfected with G5B-luciferase, GAL4-CREB and RSV- β gal plasmids (control transfection) is arbitrarily shown as 100. Results are mean $\pm$ s.d. (n =3).

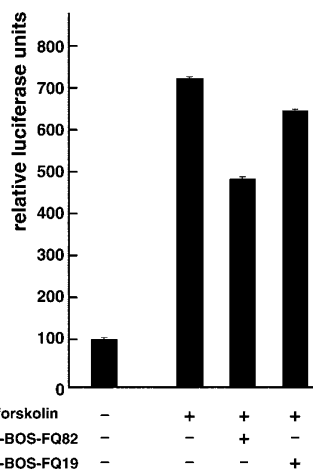
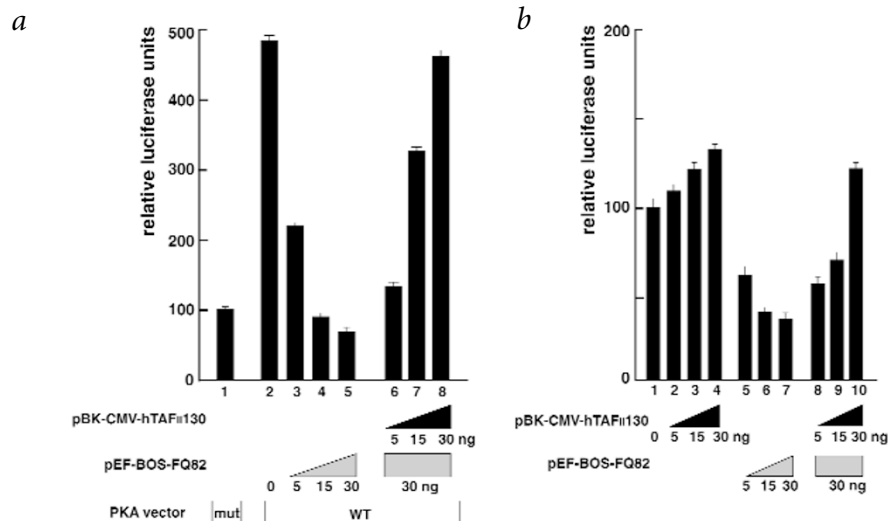


Fig. 6 TAF_{II}130 restores CREB-dependent transcriptional activation inhibited by an expanded polyQ stretch (Q82). **a**, To determine the effect of TAF_{II}130 on CREB-dependent transcriptional activation, various amounts of pEF-BOS-FQ82 (0, 5, 15, 30 ng; lanes 2–5) and pBK-CMV-hTAF_{II}130, which was further co-transfected (5, 15, 30 ng; lanes 6–8), were co-transfected into 293 cells along with a G5B-luciferase reporter plasmid, a GAL4-CREB expression vector and the wild-type (WT) PKA vector. RSV- β gal plasmid was also co-transfected to normalize the luciferase activity. The relative luciferase activity of the cells co-transfected with mutant (mut) PKA G5B-luciferase is arbitrarily shown as 100 (lane 1). Results are mean $\pm$ s.d. (n=3). **b**, To determine the effect of TAF_{II}130 on CREB-dependent transcriptional activation using the GAL4-CREB Q2 domain (GAL-Q2), various amounts of pBK-CMV-hTAF_{II}130 and pEF-BOS-FQ82 were co-transfected into 293 cells along with a G5B-luciferase reporter plasmid and the GAL-Q2 expression vector. RSV- β gal plasmid was also co-transfected to normalize the luciferase activity. The relative luciferase activity of the cells co-transfected with GAL-Q2 and G5B-Luc reporter plasmids is arbitrarily shown as 100 (lane 1). Results are mean $\pm$ s.d. (n=3). The amounts of pBK-CMV-hTAF_{II}130 and pEF-BOS-FQ82 used for the experiments are indicated below each bar. Equivalent expression concentrations of CREB, CREB-Q2 and TAF_{II}130 under all the conditions were confirmed by western-blot analysis (data not shown).



expanded polyQ stretches (Q57). Co-expression of TAF_{II}130 significantly reduced the percentage of cell death from $41.0 \pm 3.4\%$ to $23.4 \pm 2.3\%$ ($P < 0.05$; Fig. 7c). There was no significant reduction in the rate of aggregate body formation (Fig. 7d).

Discussion

We identified TAF_{II}130 as a protein that interacts with mutant proteins containing expanded polyQ stretches. The binding was confirmed by four independent methods, including the yeast two-hybrid, far-western, co-precipitation and co-localization assays. These observations are further supported by the co-localization of TAF_{II}130 *in vivo* with N1s in the autopsied brains of patients with DRPLA and MJD/SCA3. Because there are no common homologous domains among these truncated proteins used in the transient expression studies except for the polyQ stretches, and the regions flanking the polyQ stretches in the truncated DRPLA protein and ataxin-2 are particularly short (the numbers of amino acid residues upstream and downstream were 8 and 8 for truncated DRPLA protein, and 8 and 9 for the truncated ataxin-2, respectively), it is suggested that TAF_{II}130 interacts with the polyQ stretch itself. The length-dependency of the binding of polyQ stretches with TAF_{II}130 further supports this interpretation.

Several proteins have been identified to bind to the gene products of CAG repeat diseases²⁷, including HAP1 (ref. 28), HIP1 (ref. 29), LANP (ref. 30), intermediate filaments³¹ and PQBP-1 (ref. 32). In contrast to the finding that most of these proteins do not directly bind to the expanded polyQ stretches, PQBP-1 has recently been shown to interact preferentially with expanded polyQ stretches³². Although PQBP-1 inhibits transcriptional activation of a brain-specific transcription factor, Brn-2, the role of the interaction between PQBP-1 and expanded polyQ stretches in the pathogenesis of CAG repeat diseases remains to be investigated.

TAF_{II}130 is a transcriptional coactivator associated with TBP, which is required for CREB-dependent transcriptional activation. CREB mediates transcriptional activation of several genes that contain cAMP-responsive elements (CREs) through PKA-mediated phosphorylation of CREB at Ser-133 (ref. 33). The activation domain of CREB consists of a glutamine-rich Q2 constitutive domain and a kinase-inducible domain²⁶ (KID). For target gene activation, a phospho-(Ser-133)-dependent interaction between

KID and RNA polymerase II via CBP, and interaction between the Q2 domain and TFIID via TAF_{II}130 are required²⁵. Our results indicate the possibility that the interaction between TAF_{II}130 and expanded polyQ stretches interferes with CREB-dependent transcriptional activation. This possibility is supported by the observation that expanded polyQ stretches inhibited CREB-dependent transcriptional activation. Increase in luciferase activities in cells co-expressing expanded polyQ stretches and TAF_{II}130 is counter to the interpretation that the inhibition of CREB-dependent transcriptional activation was a non-specific effect associated with the strong cytotoxicity of expanded polyQ stretches.

There have been reports suggesting the possibility that CREB-dependent transcriptional activation is altered in CAG-repeat diseases. First, the expression concentrations of cAMP-responsive genes, including those encoding enkephalin, somatostatin and CRH, have been shown to be selectively decreased in CAG-repeat diseases^{34–36}. Second, reduction in long-term potentiation (LTP) in hippocampal neurons has recently been demonstrated in a 'knock-in' mouse model of HD (ref. 37). Third, stably transfected PC12 cells that express the N-terminal fragment of mutant huntingtin lack the neurite extension response to nerve growth factor³⁸. These findings could be explained by the inhibition of CREB-dependent transcriptional activation.

One of the most noteworthy findings of this study is that expanded polyQ-induced death of COS-7 and neuro2a cells was suppressed by TAF_{II}130. Recent studies have shown that mutant CREB reduced the survival of cerebellar granule cells, and that CREB is a critical regulator of neurotrophin-dependent neuronal survival partly through *bcl-2* transcription^{39,40}. Moreover, cAMP has recently been shown to partially suppress polyQ-induced cytotoxicity in primary rat cerebellar granule neurons⁴¹, in which increased intracellular cAMP concentrations may have stimulated the phosphorylation of CREB by PKA.

In addition to CREB, TAF_{II}130 has also been shown to directly bind to Sp1 (refs 21,25). Although Sp1-dependent transcriptional activation was not examined in here, it would be intriguing to investigate whether the binding of expanded polyQ stretches to TAF_{II}130 also interferes with Sp1-dependent transcriptional activation.

In the immunohistochemical observation of autopsied brains of DRPLA and MJD/SCA3 cases, we found that TBP, CREB and

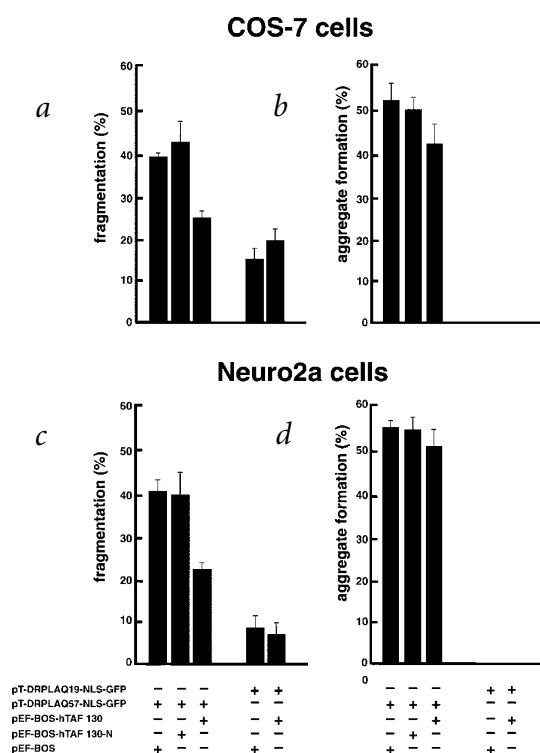


Fig. 7 Suppression of expanded polyQ stretch-induced cytotoxicity by TAF_{II}130. COS-7 cells (**a,b**) or Neuro2a cells (**c,d**) were transfected with plasmids coding for truncated DRPLA proteins with polyQ stretches, pT-DRPLAQ19-NLS-GFP or pT-DRPLAQ57-NLS-GFP, and analysed for nuclear fragmentation (**a,c**) or aggregate formation (**b,d**). To determine the effect of TAF_{II}130 on nuclear fragmentation or aggregate formation induced by the expression of truncated DRPLA protein with an expanded polyQ stretch (Q57), cells were co-transfected with pEF-BOS-hTAF_{II}130 coding for full-length TAF_{II}130, pEF-BOS-hTAF_{II}130-N coding for the N-terminal three fourths of TAF_{II}130 or an empty vector (pEF-BOS). Nuclear fragmentation was quantitated by staining cells with Hoechst 33342 (5 μ M) 48 h after the transfection. When the cells were transfected with pT-DRPLAQ19-NLS-GFP, aggregate bodies were not formed. Values are expressed as the mean $\pm$ s.d. (n=5).

Our results have demonstrated that TAF_{II}130 interacts preferentially with expanded polyQ stretches, and that this interaction inhibits CREB-dependent transcriptional activation. Suppression of the inhibition of CREB-dependent transcriptional activation caused by expanded polyQ stretches may offer opportunities for the development of new and potent therapeutic strategies for the treatment of CAG-repeat diseases.

Methods

Yeast two-hybrid system. We subcloned DNA fragments encoding polyQ stretches⁹ into pGBT9 bait vectors (pGBT9-Q19 or pGBT9-Q82). DNA fragments encoding full-length CBP, full-length CREB, RNA helicase A or full-length TAF_{II}130 were subcloned into pGAD424 prey vectors. Yeast cells (Y190 or Y153) were transformed⁴⁵ with the lacZ reporter plasmid and the appropriate bait and prey vectors (1 μ g each), and cultured on Ura-His-Trp plates containing 2% galactose. Isolated yeast colonies were then cultured in the same liquid medium and assayed for β -galactosidase activity (Clontech). All experiments were performed in triplicate.

Co-precipitation assay. We co-transfected COS-7 cells (9.0 $\times$ 10⁶ cells) with pT-DRPLA Q57-NLS-GFP-myc- His encoding the truncated GFP-NLS-fused DRPLA protein tagged by myc and His at the C terminus and pEF-BOS-hTAF_{II}130. We obtained nuclei were obtained 60 h after transfection⁴⁶ and extracted with 200 μ l of lysis buffer (20 mM Tris-HCl, pH 7.4, containing 1% NP40 and 10% glycerol) or buffer C (20 mM HEPES, pH 7.9, 25% glycerol, 0.42 M NaCl, 1.5 mM MgCl₂, 0.5 mM dithiothreitol (DTT))⁴⁶. The nuclear extracts were combined with 5 μ l of anti-myc antibody (Invitrogen), incubated overnight at 4 $^{\circ}$ C, and then further incubated with 1.5 $\times$ 10⁸ beads of Dynabeads M-450 (sheep anti-mouse IgG; DYNAL) for 2 h at 4 $^{\circ}$ C. The beads were pulled down with a magnet (DYNAL) and washed as described³¹. The beads were re-suspended in the Laemmli sample buffer (Biorad) and boiled to prepare samples for western-blot analysis. Immunoblots were probed with the anti-hTAF_{II}p130 (4A6) monoclonal antibody (SantaCruz Biotechnology; 1:500 dilution), and developed using enhanced chemiluminescence. The nuclear extracts were also incubated with Ni-NTA magnetic agarose beads (200 μ l; Qiagen) with shaking at RT for 1 h, and subjected to western-blot analysis with the anti-hTAF_{II}p130 (4A6) monoclonal antibody as described above.

Far-western-blot analysis. We generated GST-fusion proteins containing aa 1–365, 236–526, 440–670 and 577–947 of TAF_{II}130 (ref. 47). The proteins were then separated by SDS-PAGE and transferred to a nitrocellulose membrane. The membrane was soaked in a blocking solution (phosphate-buffered saline (PBS) containing 0.1% Triton X-100, 1% BSA and 5% non-fat dried milk) with gentle shaking for 1 h. After rinsing with buffer A (10 mM Tris-HCl, pH 7.5, containing 100 mM NaCl, 5 mM MgCl₂, 1 mM DTT, 0.5% BSA, 0.5% non-fat dried milk, 0.1% Triton X-100, 1 mM EDTA and 5% glycerol), the membrane was incubated in buffer A containing 10⁵ cpm/ml of a ³⁵S-labelled polyQ probe (Q82) for 2 h, washed twice with buffer A and autoradiographed. Labelling of polyQ (Q82) peptide with ³⁵S was performed using a T7 *in vitro* transcription/translation system (Promega).

Plasmid construction. The plasmid, pT-DRPLAQ56-GFP, has been described³¹. pT-DRPLAQ57-NLS-GFP was generated by PCR using pT-DRPLAQ56-GFP as the template, followed by subcloning into pEGFPN1

Sp1, in addition to TAF_{II}130, were also co-localized with NIIs. Co-localization of TBP with NIIs has also been reported in MJD/SCA3 brain⁴². Co-localization of CBP with aggregate bodies of expanded polyQ stretches has also been demonstrated in a cell culture system⁴³. These results indicate that sequestration of these nuclear proteins into NIIs has an important role in the pathogenesis of CAG-repeat diseases.

The role of aggregate formation in CAG-repeat diseases is still under debate^{19,20}. It would be intriguing to investigate whether the association of TAF_{II}130 with expanded polyQ stretches before aggregate body formation alters the transcriptional regulation, or, alternatively, whether formation of aggregate bodies containing the expanded polyQ stretches and TAF_{II}130 is a prerequisite for the toxicity.

Based on northern-blot analysis of human tissues, TAF_{II}130 is expressed ubiquitously, including the central nervous system (data not shown). Detailed analysis on the regional expression patterns of TAF_{II}130 will be required to determine whether the selective distribution of neurodegeneration in CAG-repeat diseases is related to regional differences in the expression concentrations of TAF_{II}130.

As revealed by the yeast two-hybrid assay and CREB-dependent transcriptional activation assay, even a short polyQ stretch (Q19) was bound to TAF_{II}130 and suppressed the CREB-dependent transcriptional activation, but to a weaker extent than the expanded polyQ stretch (Q82). This might argue against the role of the binding in the pathogenesis of CAG-repeat diseases, because threshold lengths of the polyQ stretches have been demonstrated. To address this issue, detailed analysis of the binding of TAF_{II}130 to polyQ stretches of various lengths and the effect of these polyQ stretches on transcriptional activations mediated by TAF_{II}130 will be required. Our results also raise the possibility that polyQ stretches even within the normal lengths may have some effects on TAF_{II}130-mediated transcription by their interaction with TAF_{II}130 under physiological conditions. Because many nuclear proteins are known to contain various lengths of polyQ stretches⁴⁴, this issue needs to be investigated further.

(Clontech). pT-DRPLAQ19-NLS-GFP, pT-HDQ81-GFP, pT-HDQ82-NLS-GFP, pT-SCA2Q64-GFP and pT-MJDQ72-GFP were similarly constructed using PCR. pT-DRPLAQ57-NLS-GFP-myc-His was constructed by ligating the *XhoI/BamHI* segment of pT-DRPLAQ57-NLS-GFP into pcDNA3.1/myc-His (Invitrogen). We constructed pEF-BOS-hTAF_{II}130 by ligating the *Sse8387I/NotI* segment of pBK-CMV-hTAF_{II}130 (ref. 21) into pEF-BOS (ref. 48). pEF-BOS-hTAF_{II}130-N, encoding aa 1–682 of TAF_{II}130, was also generated by a PCR-based method. The nucleotide sequences of all the plasmids were confirmed using automated DNA sequencers (PE Applied Biosystems).

Immunofluorescence and confocal microscopy. COS-7 cells seeded onto an 8-well chamber slide (Nunc) were transfected with plasmid DNA (0.25 µg each) using the SuperFect transfection reagent (Qiagen) according to the manufacturer's instructions. The cells were fixed in 0.1 M PBS containing 4% paraformaldehyde for 30 min, permeabilized with PBS containing 0.2% Triton X-100 for 30 min and incubated in a blocking solution (PBS containing 10% goat serum and 0.02% Triton X-100) for 60 min at RT. Cells were incubated for 2 h at RT with anti-hTAF_{II}p130 (4A6) monoclonal antibody (1:500 dilution) diluted in the blocking solution, and then with Cy5-conjugated anti-mouse IgG (Amersham; 1:100) for 1 h at RT. Confocal images were obtained using the MRC-1024 Confocal System (Biorad).

Immunohistochemistry. Autopsied brains of three DRPLA cases, N17 (88; female, Q55, onset 61 y, duration 15 y), N20 (91; female, Q64, onset 10 y, duration 18 y) and N8 (97; female, Q67, onset 16 y, duration 13 y) and a MJD/SCA3 case, N24 (86; female, Q83, onset 14 y, duration 18 y) were subjected to immunohistochemical analysis. Tissue specimens were obtained at autopsy, quick-frozen in cold isopentane and stored in a deep freezer until use. Cryostat sections (8 µm) were made from the frozen materials, fixed with cold acetone (–20 °C) for 7 min and immunostained by the avidin-biotin-peroxidase complex (ABC) method using a Vectastain ABC kit (Vector) with antibodies against hTAF_{II}p130 (4A6, 1:100), TFIID (TBP; SI-1, 1:2,000), CREB-1 (24H4B, 1:100), Sp1 (PEP 2, 1:4,000), C/EBP-α (14AA, 1:2,000), C/EBP-β (C-19, 1:2,000) and C/EBP-β (Δ198, 1:2,000; Santa Cruz Biotechnology). The sections were counterstained with haematoxylin.

Assay of CREB-dependent transcriptional activation. Six-well culture dishes were seeded with 293 cells, and transfection assays were performed using the Lipofectin reagent (Gibco-BRL). A G5B-luciferase reporter plasmid (500 ng) containing 5 GAL4 recognition sites was co-transfected with

a GAL4-CREB expression vector (500 ng), a wild-type PKA vector²⁶ (10 ng), RSV-β gal control plasmid (20 ng) and/or a polyQ expression plasmid (100 ng; pEF-BOS-FQ19 or pEF-BOS-FQ82). A mutant PKA vector coding for catalytically inactive PKA with a Lys/Met72 substitution in the ATP-binding site²⁶ was used as a control for the wild-type PKA. Six hours after transfection, forskolin was added into the medium. After 24 h, the cells were collected, lysed with a cell lysis buffer (Toyo Ink) and assayed for luciferase activity using an Auto-Lumat Luminometer (Berthold). Luciferase activity was normalized to the β-galactosidase activity. All experiments were performed in triplicate. A plasmid for the constitutive Q2 domain fused to GAL4 (GAL-Q2 protein) was also used in place of GAL4-CREB and PKA vectors for evaluation of the transcriptional activation of the G5B-luciferase reporter gene.

Quantitative assay for cell death. As a measure of cell death, nuclear fragmentation was quantitated as described⁹. Statistical analyses were performed using Student's *t*-test.

GenBank accession numbers. CBP, S66385; CREB, X14788; RNA helicase A, Y10658.

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